

Anchor Datasets

CSAI2017P: Applied Machine Learning (Lab)

Reference for lab assignments LA1–LA10

The ten lab assignments reuse three families of dataset. The point is that you learn the methods rather than spending every fortnight wrestling a new file format. Get each one *once* into a local `data/` folder and keep it there for the semester.

Do this at home, before the lab

Only some of these ship inside scikit-learn. The rest **download over the internet on first use**, and the campus network may block or intercept that download. If you first run the code in the lab, you may lose the entire session. **Run the pre-fetch script below on your own connection, at home, before the lab it is needed for.** Once fetched, the data is cached on your machine and works offline afterwards.

Anchor A – Tabular records

Mixed numeric and categorical rows, one record per entity. Used for EDA, regression, classification, scoring and anomaly work.

ID	Dataset	Source	Notes
A1	California Housing	<code>sklearn.datasets.fetch-california_housing</code>	20,640 rows, 8 numeric features, numeric target MedHouseVal (\$100k units). Downloads on first use (~1.4 MB from figshare), then caches. Fetch it at home before the lab — see <i>Getting the data</i> below.
A2	Telco Customer Churn	Kaggle: <code>blastchar/telco-customer-churn</code>	7,043 rows, 21 mixed features, binary target Churn. Download the CSV once and keep it in <code>data/</code> .
A3	Statlog German Credit	OpenML: <code>credit-g</code> (id 31)	1,000 rows, 20 mixed features, binary target class (good/bad credit). Downloads on first use via <code>fetch_openml('credit-g', version=1)</code> , then caches. Fetch it at home.

Anchor B – Images and image-derived features

Pixel grids and features measured from images. Used for image classification, clustering on images and diagnostic tasks.

ID	Dataset	Source	Notes
B1	scikit-learn digits	<code>sklearn.datasets.load_digits</code>	1,797 images of 8×8 greyscale handwritten digits, 10 classes. Bundled with scikit-learn — works offline. A small stand-in for MNIST that trains in seconds.
B2	MNIST (full)	<code>sklearn.datasets.fetch_openml('mnist_784')</code>	70,000 images of 28×28 greyscale digits. Downloads on first use (~15 MB), then caches. Use a subsample of 10,000 unless you have time to spare.
B3	Breast Cancer Wisconsin (Diagnostic)	<code>sklearn.datasets.load_breast_cancer</code>	569 rows, 30 features computed from digitised images of fine-needle aspirates, binary target malignant/benign. Bundled with scikit-learn — works offline.
B4	PlantVillage (leaf subset)	Kaggle: <code>emmarex/plantdisease</code>	Leaf photographs labelled by crop and disease. Use the tomato subset (10 classes) resized to 64×64 to keep runtimes sane.

Anchor C – Sequences and text

Data with an order that matters, or free text. Used for time-series regression, text classification and signal work.

ID	Dataset	Source	Notes
C1	Daily stock OHLCV	Kaggle: <code>any single-ticker daily history, or yfinance</code>	One row per trading day with Open/High/Low/Close/Volume. Pick one ticker and at least three years of history.
C2	SMS Spam Collection	UCI ML Repository: <code>SMS Spam Collection</code>	5,574 English SMS messages labelled ham/spam. A single tab-separated file.
C3	UCI Human Activity Recognition	UCI: <code>Human Activity Recognition Using Smartphones</code>	561 features engineered from accelerometer and gyroscope windows, 6 activity classes, subject-wise train/test split supplied.

Loading snippets

Getting the data

Bundled with scikit-learn — nothing to do, works offline: `load_digits`, `load_breast_cancer`, `load_iris`, `load_diabetes`.

Downloads once, then cached — run this at home before the lab:

```
# Pre-fetch script. Run ONCE at home, on a normal internet connection.
from sklearn.datasets import fetch_california_housing, fetch_openml

fetch_california_housing() # A1 ~1.4 MB
fetch_openml("credit-g", version=1, as_frame=True) # A3 ~0.2 MB
# fetch_openml("mnist_784", version=1, as_frame=True) # B2 ~15 MB, only if LA10 needs it
print("cached OK")
```

The cache lives in `~/scikit_learn_data/` (`C:\Users\<you>\scikit_learn_data` on Windows).

If the download fails on the campus network you will see `CERTIFICATE_VERIFY_FAILED`. That is the campus web filter, not your code, and you cannot fix it by editing Python. Three things that do work, best first:

1. Run the pre-fetch script above **at home**, on a normal connection. The cache then persists and everything works offline afterwards.
2. **Download it in your browser** — browsers accept the campus certificate even when Python does not — and save the file into the cache folder above.
3. Copy the whole `scikit_learn_data` folder from a classmate who already has it. This is allowed, and is often the fastest fix in the room.

Never do this

Do not *fix* the error by turning off certificate verification. It teaches you nothing, it will not be accepted in your submission, and it is a habit that causes real harm outside this course.

Manual download — fetch the file yourself and put it in `data/`:

```
# A1 California Housing (downloads on first use, then cached)
from sklearn.datasets import fetch_california_housing
X, y = fetch_california_housing(return_X_y=True, as_frame=True)

# A3 German Credit (downloads on first use, then cached)
from sklearn.datasets import fetch_openml
d = fetch_openml("credit-g", version=1, as_frame=True)

# B1 digits (bundled, offline)
from sklearn.datasets import load_digits
dig = load_digits() # dig.images 8x8, dig.data 64 features

# B3 Breast Cancer Wisconsin (bundled, offline)
from sklearn.datasets import load_breast_cancer
bc = load_breast_cancer(as_frame=True)

# A2 Telco / C2 SMS Spam / C1 stock / B4 leaves : place the CSV in data/
import pandas as pd
df = pd.read_csv("data/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

Reproducibility

Record the exact file name, the download date and the row count of every downloaded dataset in your notebook. Public datasets are revised without notice, and a result you cannot reproduce is a result you cannot defend in the lab quiz.

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.; McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.; Scikit-learn user guide, accessed 2026-08-04.; Matplotlib and Seaborn documentation, accessed 2026-08-04.